

Prime Oscillation and the Explicit Formula: A Rigorous Framework for Zeta Zero Reconstruction

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Abstract

We reinterpret the Prime Eigenvalue Oscillation Hypothesis (PEOH) within the rigorous framework of the Weil explicit formula. By expressing smoothed zero-density data as a superposition of von Mangoldt-weighted prime-power harmonics, we obtain an exact finite reconstruction identity for band-limited test functions. We formalize the bandwidth-resolution-cost tradeoff for zero reconstruction, propose a reproducible numerical validation protocol, and state a trace-realization conjecture that cleanly separates what is rigorous now from what would constitute RH-level progress. This is **not** a proof of the Riemann Hypothesis (RH); rather, it is a mathematically defensible research program connecting explicit formula technology with spectral/trace interpretations.

1. Introduction

The Riemann Hypothesis (RH) asserts that all nontrivial zeros of $\zeta(s)$ lie on $\Re(s) = \frac{1}{2}$. Attempts to “explain” zeros as interference among prime oscillations are common but often informal. This note salvages the prime-oscillation intuition by placing it squarely inside the Weil/Guinand explicit formula. In this framework, a smoothed zero measure is exactly equal (for suitable band-limited kernels) to a computable Archimedean term plus a **finite** von Mangoldt-weighted oscillatory sum over $n \leq e^L$ (hence including prime powers).

Contributions.

1. RH-neutral formalization of “prime oscillation $\leftrightarrow$ zeros” via the shifted explicit formula.
2. Exact finiteness of the prime-power side for compactly supported $\hat{f}$.
3. A bandwidth-resolution-cost tradeoff: resolving zeros near height T requires $L \gg \log T$ and prime powers up to scale $e^L \gtrsim T$.
4. A reproducible numerical protocol to verify the explicit formula numerically and to build a zero-locating algorithm.
5. A logically consistent trace-realization conjecture (RH-strength) plus a conditional positivity sanity check.

Related work (high-level)

- Weil/Guinand explicit formula and test-function formalism.
- Selberg trace formula as a guiding analogy.
- Connes’ trace-formula viewpoint in noncommutative geometry.
- Berry–Keating semiclassical operator heuristics.
- Montgomery pair correlation and Odlyzko’s computations of zero statistics.

2. Mathematical Framework

2.1 Fourier convention

We fix the normalization

$$\widehat{f}(u) = \int_{-\infty}^{\infty} f(t) e^{-iut} dt, \quad f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \widehat{f}(u) e^{iut} du.$$

2.2 Completed zeta

Let

$$\xi(s) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma(s/2) \zeta(s).$$

Then ξ is entire and satisfies $\xi(s) = \xi(1-s)$.

2.3 Zeros and an RH-neutral spectral parameter

Let nontrivial zeros of ζ be $\rho = \beta + i\gamma$ with multiplicity. Define

$$\theta_\rho := \frac{\rho - \frac{1}{2}}{i} = \gamma + i \Big(\frac{1}{2} - \beta \Big) \in \mathbb{C}.$$

Under RH, $\beta = \frac{1}{2}$ and $\theta_\rho = \gamma \in \mathbb{R}$.

2.4 Test-function class

Assume

- $\hat{f} \in C_0^\infty([-L, L])$ is even and real-valued;
- f is defined via the inverse Fourier transform.

Then f is even, real, Schwartz on $\mathbb{R}$, and entire (Paley–Wiener).

3. Shifted bandlimited explicit formula

Define the smoothed zero-density (RH-neutral):

$$Z_f(t) := \sum_{\rho} f \big(t - \theta_\rho \big),$$

where the sum is over nontrivial zeros with multiplicity.

Proposition 1 (Shifted bandlimited explicit formula)

There exists an explicit Archimedean term

$$A_f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(t-\tau) \Psi(\tau) d\tau,$$

where

$$\Psi(\tau) = \frac{\Gamma'(\frac{1}{2})}{\Gamma(\frac{1}{2})} \text{Big}(\frac{1}{4} + \frac{i\tau}{2}) + \frac{\Gamma'(\frac{1}{2})}{\Gamma(\frac{1}{2})} \text{Big}(\frac{1}{4} - \frac{i\tau}{2}) - \log \pi,$$

such that

$$Z_f(t) = A_f(t) - \frac{1}{2} \sum_{n \leq e^L} \frac{\Lambda(n)}{\sqrt{n}} \widehat{f}(\log n) \cos(t \log n).$$

Finiteness. Since $\text{supp } \hat{f} \subset [-L, L]$, we have $\hat{f}(\log n) = 0$ for $n > e^L$. Hence the von Mangoldt sum is finite and exact at the truncation level $n \leq e^L$.

Reality. For real-even f , the symmetry $\rho \mapsto \bar{\rho}$ implies $Z_f(t) \in \mathbb{R}$ for real t .

RH specialization. If RH holds, $\theta_\rho = \gamma \in \mathbb{R}$ and $Z_f(t) = \sum_\gamma f(t - \gamma)$.

Remark (shift mechanism). For the shifted kernel $f_t(x) := f(t - x)$, one has $\widehat{f_t}(u) = e^{-iut} \hat{f}(u)$, yielding cosine terms for real-even f .

4. Prime oscillation interpretation

Define the prime-power oscillation term

$$P_f(t) := -2 \sum_{n \leq e^L} \frac{\Lambda(n)}{\sqrt{n}} \widehat{f}(\log n) \cos(t \log n).$$

Then Proposition 1 reads $Z_f(t) = A_f(t) + P_f(t)$.

Interpretation. The smoothed zero measure Z_f is exactly the superposition of:

1. a smooth Archimedean density term A_f , and 2) a finite interference pattern of prime-power harmonics P_f .

Zero reconstruction mechanism (conditional)

If $f \geq 0$ is unimodal and sharply peaked at 0, then Z_f is a superposition of bumps centered at θ_ρ . Under RH, these are centered at the ordinates γ , making local maxima of $Z_f(t)$ (equivalently, minima of $P_f(t)$ after subtracting $A_f(t)$) natural zero estimators.

5. Bandwidth–resolution–cost tradeoff

Let f_L be a family with $\text{supp } \hat{f}_L \subset [-L, L]$.

Proposition 2 (Bandwidth–resolution principle)

The effective temporal width satisfies $\Delta t \gtrsim 1/L$. To resolve individual zeros near height T , whose mean spacing is approximately

$$\Delta\gamma(T) \sim \frac{2\pi}{\log(T/2\pi)},$$

one requires $\Delta t \ll \Delta\gamma(T)$, hence

$$L \gg \log T.$$

Proposition 3 (Bandwidth–cost tradeoff)

Because $\hat{f}(\log n)$ vanishes for $n > e^L$, reconstructing Z_f at bandwidth L requires von Mangoldt data up to $n \leq e^L$, i.e.

$$\text{prime-power cutoff} \asymp e^L.$$

In particular, resolving zeros at height T at the mean-spacing scale requires $e^L \gtrsim T$, and computational cost is dominated by generating $\Lambda(n)$ up to N .

6. Numerical validation program

6.1 Deliverable A: explicit-formula verification

Kernel construction. Choose $\hat{f}_L \in C_0^\infty([-L, L])$, even and real; define f_L by inverse Fourier transform.

Compute three components independently.

1. $A_{f_L}(t)$: numerical quadrature of the Ψ -integral.
2. $P_{f_L}(t)$: finite von Mangoldt sum over $n \leq e^L$ (prime powers included).
3. $Z_{f_L}(t)$: from tabulated zeros ρ (or ordinates γ under RH) up to cutoff $T_{\max}$.

Verify

$$Z_{f_L}(t) \approx A_{f_L}(t) + P_{f_L}(t)$$

on a window $t \in [T_1, T_2]$, and record

$$\epsilon(L, T_{\max}) = \sup_{t \in [T_1, T_2]} \left| Z_{f_L}^{\{\text{zeros}\}}(t) - \big(A_{f_L}(t) + P_{f_L}(t)\big) \right|.$$

Error controls.

- *Prime side:* exact at truncation level $n \leq e^L$; numerical error from floating-point evaluation of $\cos(t \log n)$.
- *Zero-side truncation:* for Schwartz f_L , a safe tail control uses the zero counting measure dN :

$$\sum_{|\gamma| > T_{\max}} |f_L(t - \gamma)| \lesssim \int_{T_{\max}}^{\infty} |f_L(u)| \log u \, du.$$

- *Quadrature error:* controlled via adaptive integration.

6.2 Deliverable B: zero location accuracy

Detection kernels (relaxed smoothness). Fejér-type kernels are band-limited but not C^∞ :

$$\widehat{f}_L(x) = \max(1 - |x|/L, 0).$$

These often yield robust bump-like reconstructions for detection.

Protocol.

1. Compute $Z_{f_L}(t)$ or $P_{f_L}(t)$ on $[T_1, T_2]$.
2. Detect candidate ordinates via maxima of Z_{f_L} (or minima of P_{f_L} after subtracting A_{f_L}).
3. Compare to known ordinates γ_n and measure mean/max error and detection rate.

7. Operator-theoretic outlook

7.1 Conjecture (self-adjoint trace realization; RH-strength)

Let $W(f)$ denote the Weil explicit-formula functional (the right-hand side of Proposition 1 evaluated at $t = 0$, or the corresponding distributional functional).

Conjecture 1. There exists a Hilbert space $\mathcal{H}$, a self-adjoint operator H with discrete spectrum $\{\gamma_n\} \subset \mathbb{R}$, and a trace regularization $\mathrm{Tr}_{\mathrm{reg}}$ such that for all even f with compactly supported $\hat{f}$,

$$\mathrm{Tr}_{\mathrm{reg}}(f(H)) = W(f).$$

This is RH-strength: a self-adjoint spectral realization would force the relevant spectral parameters to be real.

7.2 Conditional positivity sanity check

Proposition 4 (conditional positivity). Assume Conjecture 1 holds and $\mathrm{Tr}_{\mathrm{reg}}$ agrees with the usual trace on positive trace-class operators. Then for all $f \geq 0$ in the test class for which $f(H)$ is trace-class,

$$\mathrm{Tr}(f(H)) \geq 0.$$

This yields a necessary-condition numerical sanity check (dependent on the regularization).

7.3 Candidate operator: Berry–Keating with cutoff

Consider $H_{\mathrm{BK}} = \frac{1}{2}(xp + px)$ with boundary/cutoff choices designed to yield a discrete or effectively discrete spectrum. The research goal is to establish a controlled semiclassical trace approximation matching $W(f)$ at leading order and to bound the remainder under a specific cutoff/regularization.

8. Conclusion

The prime-oscillation intuition can be stated rigorously: for band-limited test functions, a smoothed zero-density is exactly the sum of an Archimedean gamma term and a finite von Mangoldt-weighted prime-power interference pattern. This gives a principled route to zero reconstruction with explicit bandwidth-resolution-cost tradeoffs and a clean numerical validation protocol. The operator-theoretic extension is explicitly RH-strength and is therefore framed as a conjectural pathway rather than a proof.

Appendix A. Kernel construction examples

A standard smooth compactly supported bump:

$$\widehat{f}_L(x) = \begin{cases} \exp\left(-\frac{1}{1-(x/L)^2}\right), & |x| < L, \\ 0, & |x| \geq L. \end{cases}$$

Then define f_L by inverse Fourier transform.

Detection kernels with relaxed smoothness (Fejér-type):

$$\widehat{f}_L(x) = \max(1 - |x|/L, 0).$$

References

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